

TEST PLAN

IIITD ERP System

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1. Introduction

This Test Plan describes the functional and non-functional tests performed on the IIITD ERP System.

The goal is to verify:

- Correct functioning of Student, Instructor, and Admin flows
- Enforcement of access rules, maintenance mode, and validation
- Authentication logic and the separation of Auth DB and ERP DB
- Stability and correctness of all UI screens

The app is tested using the seeded accounts:

Username	Role	Notes
admin1	Admin	From seed SQL
inst1	Instructor	Assigned at least one section
stu1	Student	Unenrolled or minimal data
stu2	Student	Enrolled in at least one section

2. Test Environment

- **OS:** Windows 11
- **Java Version:** 17

- **MySQL Version:** 8.x server running locally
 - **Maven Version:** 3.6+ (tested with 3.9.x)
 - **Databases:**
 - `auth_db` (Auth DB) – created via seed script
 - `erp_db` (ERP DB) – created via seed script
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3. Test Data Used

Mention course IDs, section IDs, and any seen sample values:

Example:

- Course: `CSE101` — Intro to Programming
- Section: `section_id = 1`, capacity 40
- Instructor assigned: `inst1`
- Student `stu2` enrolled in this section

(Replace with your actual values.)

4. Acceptance Tests (from specification)

Rewrite acceptance tests in your own words.

4.1 Login Tests

Test ID	Description	Steps	Expected Result
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L1	Wrong password rejected	Login as stu1 with wrong password	"Incorrect username or password."
L2	Correct dashboard loads	Login as admin1	Admin dashboard appears

4.2 Student Tests

Test ID	Description	Steps	Expected Result
S1	View catalog	stu1 → Catalog	Table shows code/title/credits/slots
S2	Register section success	Choose open section → Register	Success message + appears in My Registrations
S3	Duplicate registration blocked	Register again same section	"Already registered"
S4	Full section blocked	Register full section	"Section full"
S5	Drop before deadline	My Registrations → Drop	Enrollment removed
S6	View grades	stu2 → Grades	Components + final grade visible
S7	Transcript export	stu2 → Export Transcript	CSV/PDF downloads

4.3 Instructor Tests

Test ID	Description	Steps	Expected Result
I1	View only own sections	inst1 → My Sections	Only assigned sections appear
I2	Enter scores	Open section → Edit cells → Save	Scores saved
I3	Compute final grade	Compute Final → weights entered	Final grade computed & saved

4.4 Admin Tests

Test ID	Description	Steps	Expected Result
A1	Create user	Add User → Submit	User appears in list + Auth DB entry
A2	Create course & section	Create Course/Section	Values stored
A3	Assign instructor	Assign inst1 to section	Stored correctly
A4	Toggle maintenance	ON → Banner appears	Writes blocked for students/instructors

4.5 Security + Data Separation

Test ID	Description	Steps	Expected Result
SEC1	Auth DB only stores hashes	View users_auth	No plaintext passwords
SEC2	ERP DB has no passwords	View ERP tables	No password columns
SEC3	Login loads profile from ERP DB	Login as stu1	student.user_id matches users_auth.user_id

4.6 Edge Case Tests

Test ID	Description	Expected Result
E1	Negative capacity rejected	Error shown
E2	Register after deadline	Rejected with message
E3	Duplicate enrollments prevented	UNIQUE constraint enforced

Summary:

Summary Table

Test ID	Test Description	Result (Pass/Fail)	Notes
L1	Wrong password rejected	PASS	
L2	Role dashboard loads	PASS	
S1	Student catalog loads	PASS	
S2	Registration success	PASS	
S3	Duplicate registration	PASS	
S4	Full section block	PASS	
S5	Drop section before deadline	PASS	
S6	View grades	PASS	
S7	Transcript export	PASS	
I1	Instructor sees only own sections	PASS	
I2	Enter scores	PASS	
I3	Compute final grade	PASS	
A1	Create user	PASS	
A2	Create course & section	PASS	
A3	Assign instructor	PASS	
A4	Toggle maintenance	PASS	
SEC1	Password hashing	PASS	
E1	Negative capacity rejected	PASS	

E3	Duplicate enrollment constraint	PASS
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(Add/remove tests as needed depending on your actual coverage.)